WolfBite

For teams choosing a Project 3 codebase

A WIC shopping assistant that explains what to do when checkout says no

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The problem and who benefits

The problem: WIC participants scan an item, but the app shows it as eligible while the register rejects it. This causes confusion, frustration, and delays.

Who benefits: Caregivers shopping with young children, WIC clinic staff who answer questions, and cashiers who face disputes at checkout.

Our gap: Existing apps show benefits and allow scanning, but do not give a clear, item-level explanation and next steps when an item is rejected.

What WolfBite does today

- **Scan** a **barcode** or enter a UPC and check it against the real NC WIC Authorized Product List (APL).
- Keep a shopping basket with category allowances. Extra items show as shopper-paid.
- View benefit balances, save sessions, and reset monthly.
- Show nutrition information and healthier alternatives.
- If an item is rejected at checkout, give likely reasons and next steps, or say "unable to determine".
- Sign in with Firebase and store data in Firestore.

Important limits

WolfBite is not the WIC system. It only shows possible reasons and next steps. The register is always right.

Allowances are demo values, not live benefit balances. Checkout scenarios are mock cases.

If a product is not in the APL, it stops at "not found" instead of reaching checkout help.

Project 2: the five milestones we built

M0 — Evaluate checkout help

Tested whether the checkout-help feature is useful. Users were given scenarios with and without help. Target: 80% picked the right next step in 60 seconds.

M1 — Nutrition explainers

Show nutrition information, serving size, and a plain-language note per nutrient. If data is missing, show "Unknown" and do not give a badge.

M2 — Basket-wide suggestions

Add a "Basket balance" mode that shows one-swap suggestions for the whole basket, using the same benefit category only.

M3 — Clearer interface

Improve and complete the shopping interface. Consistent scan, basket and benefits screens, with both phone and desktop layouts.

M4 — Help for a rejected item

When an item is rejected, explain possible reasons (e.g., size, balance, or outdated data) and give clear next steps, or say "unable to determine".

Project 3: four milestones your team could work on

1 — Live, versioned NC APL sync

Replace the static product file with a regularly updated APL. Show the version, state and last updated time, so users know the data is current. Needs NCDHHS data access.

2 — Manual benefit entry

Let users enter their own allowances instead of demo values. Real balances stay out of scope until a data partner exists.

3 — On-device receipt recognition

Read receipts on the device (without uploading to a cloud service) and match items to WIC products. Keep manual entry as a fallback.

4 — Spanish and accessibility validation

Add professionally reviewed Spanish text and test with screen readers, text-scaling and low-literacy users.

Why this repo is easy to extend

- **Tested: 300+** team-written tests with **93%** line coverage, plus new M1 tests.
- **Honest:** known issues are documented, not hidden.
- **Modular:** clear separation of data, logic and UI.
- **Well-structured code:** easier for a new team to understand and add features.
- **Real data:** already works with NC WIC APL.

App screenshots (current version)



Tech stack

- Flutter / Dart (mobile and web)
- Firebase Authentication and Cloud Firestore
- Provider state & GoRouter navigation
- mobile_scanner and qr_flutter
- OCR.space (receipt text)
- Mockito and flutter_test
- GitHub Actions CI
- LCOV coverage reporting

Explore WolfBite

GitHub Repository

Feature Walkthrough

Demo Video

Watch video

Team Discussion